ANNEX 3

Antenna gain towards the horizon for an earth station operating with a geostationary space station

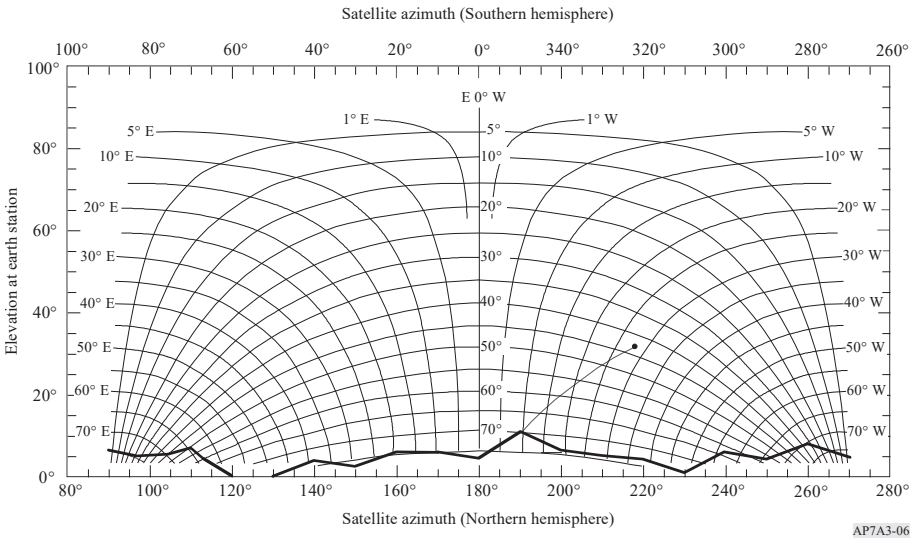
1 General

The gain component of the earth station antenna in the direction of the physical horizon around an earth station is a function of the angular separation between the antenna main beam axis and the horizon in the direction under consideration. When the earth station is used to transmit to a space station in a slightly inclined orbit, all possible pointing directions of the antenna main beam axis need to be considered. For earth station coordination, knowledge of $\varphi(\alpha)$, the minimum possible value of the angular separation that will occur during the operation of the space station, is required for each azimuth.

When a geostationary space station maintains its location close to its nominal orbital position, the earth station's main beam axis elevation angle, ε_s, and the azimuth angle, α_s, to the space station from the earth station's latitude, ζ, are uniquely related. Figure 6 shows the possible location arcs of positions of a space station on the geostationary orbit in a rectangular azimuth/elevation plot. It shows arcs corresponding to a set of earth station latitudes and the intersecting arcs correspond to points on the orbit with a fixed difference in longitude East or West of the earth station. Figure 6 also shows a portion of the horizon profile $\varepsilon_h(\alpha)$. The off-axis angle $\varphi(\alpha)$ between the horizon profile at an azimuth of 190° and a space station located 28° W of an earth station at 43° N latitude is indicated by the great-circle arc shown dashed on Fig. 6.

FIGURE 6

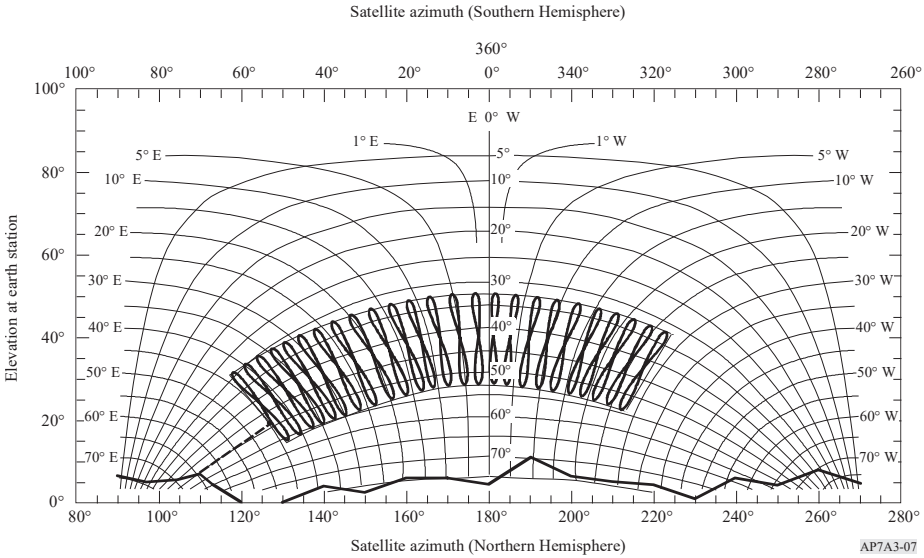
Position arcs of geostationary satellites with horizon and the arc from the horizon at azimuth 190° to a satellite 28° W of an earth station at 43° N latitude



When the north/south station-keeping of a geostationary satellite is relaxed, the orbit of the satellite becomes inclined, with an inclination that increases gradually with time. As viewed from the Earth, the position of the satellite traces a figure eight during each 24-hour period. Figure 7 shows the variations in the trajectories of a set of satellites, each with 10° inclination, spaced by 3° along the geostationary orbit from 28° W to 44° E, with respect to an earth station at 43° N latitude. Figure 7 also shows, with a dashed curve, the great-circle arc corresponding to the minimum off-axis angle φ(α) between a point on the trajectory of one of the satellites and the horizon profile at an azimuth of 110°.

FIGURE 7

Position arcs of geostationary satellites with horizon and the arc from the horizon at azimuth 110° to satellites with 10° inclination on the geostationary orbital arc from 28° W to 44° E of an earth station at 43° N latitude



For a transmitting earth station operating in a frequency band that is also allocated for bidirectional use by receiving earth stations operating with geostationary space stations, refer to § 2.1 of Annex 5.

2 Determination of the angular separation φ(α)

For the determination of the off-axis angle φ(α), two cases are distinguished. These depend on whether the orbit of the space station has no inclination, or is slightly inclined. The following equations may be used in both of these cases:

$$\psi_s(i,\delta) = \arccos\left(\sin\zeta \sin i + \cos\zeta \cos i \cos\delta\right) \tag{84}$$

$$\varepsilon_s(i,\delta) = \arcsin\left(\frac{K\cos\psi_s(i,\delta) - 1}{\left(1 + K^2 - 2K\cos\psi_s(i,\delta)\right)^{1/2}}\right) \tag{85}$$

$$\alpha_{0s}(i,\delta)=\arccos\left[\frac{\sin i-\cos\psi_s\sin\zeta}{\sin\psi_s\cos\zeta}\right] \quad (86)$$

$$\alpha_s(i,\delta)=\alpha_{0s}(i,\delta) \quad \text{for a space station located east of the earth station } (\delta \geq 0) \quad (87)$$

$$\alpha_s(i,\delta)=360°-\alpha_{0s}(i,\delta) \quad \text{for a space station located west of the earth station } (\delta \leq 0) \quad (88)$$

$$\varphi(\alpha,i,\delta)=\arccos\left[\cos\varepsilon_h(\alpha)\cos\varepsilon_s(i,\delta)\cos\left(\alpha-\alpha_s(i,\delta)\right)+\sin\varepsilon_h(\alpha)\sin\varepsilon_s(i,\delta)\right] \quad (89)$$

where:

ζ: latitude of the earth station (positive for north; negative for south)

δ: difference in longitude between the earth station and a space station

i: latitude of a sub-satellite point (positive for north; negative for south)

$\psi_s(i, \delta)$: great-circle arc between the earth station and a sub-satellite point

$\alpha_s(i, \delta)$: space station azimuth as seen from the earth station

$\varepsilon_s(i, \delta)$: space station elevation angle as seen from the earth station

$\varphi\lambda\alpha, i, \delta)$: angle between the main beam and the horizon direction corresponding to the azimuth, α, under consideration when the main beam is steered towards a space station with a sub-satellite point at latitude, i, and longitude difference, δ

α: azimuth of the direction under consideration

ε_h: elevation angle of the horizon at the azimuth under consideration, α

$\varphi(\alpha)$: angle to be used for horizon gain calculation at the azimuth under consideration, α

K: orbit radius/Earth radius, which for the geostationary orbit is assumed to be 6.62.

All arcs mentioned above are in degrees.

***Case 1*: Single space station, no orbital inclination**

For a space station operating with no orbital inclination at an orbital position with difference in longitude δ_0, equations (84) to (89) may be applied directly using $i = 0$ to determine $\varphi(\alpha)$ for each azimuth α. Thus:

$$\varphi(\alpha) = \varphi(\alpha, 0, \delta_0) \quad (90)$$

where:

δ_0: difference in longitude between the earth station and the space station.

***Case 2*: Single space station, slightly inclined orbit**

For a space station operating in a slightly inclined orbit on a portion of the geostationary arc with a nominal longitude difference of δ_0, the maximum orbital inclination over its lifetime, i_s, must be considered. Equations (84) to (89) may be applied to develop the minimum off-axis angle to each of four arcs in azimuth/elevation that bound the trajectory of the space station in angle and elevation. The bounding arcs correspond to the maximum and minimum latitudes of the sub-satellite points and the extremes of the difference in longitude between the earth and space stations when the space station is operating at its maximum inclination.

The determination of the minimum off-axis angles in equations (91) to (95) may be made by taking increments along a bounding contour. The step size in inclination *i* or longitude δ should be between 0.5° and 1.0° and the end points of the respective ranges should be included in the calculation.

The horizon profile $\varepsilon_h(\alpha)$ used in the determination of $\varphi(\alpha)$ is specified at increments in azimuth α that do not exceed 5°.

Thus:

$$\varphi(\alpha) = \min_{n = 1 \text{ to } 4} \varphi_n(\alpha) \tag{91}$$

with:

$$\varphi_1(\alpha) = \min_{\delta_0 - \delta_s \le \delta \le \delta_0 + \delta_s} \varphi(\alpha, -i_s, \delta) \tag{92}$$

$$\varphi_2(\alpha) = \min_{\delta_0 - \delta_s \le \delta \le \delta_0 + \delta_s} \varphi(\alpha, i_s, \delta) \tag{93}$$

$$\varphi_3(\alpha) = \min_{-i_s \le i \le i_s} \varphi(\alpha, i, \delta_0 - \delta_s) \tag{94}$$

$$\varphi_4(\alpha) = \min_{-i_s \le i \le i_s} \varphi(\alpha, i, \delta_0 + \delta_s) \tag{95}$$

$$\delta_s = \left(i_s / 15 \right)^2 \tag{96}$$

where:

i_s: maximum operational inclination angle of the satellite orbit

δ_s: maximum longitude change from nominal value of the sub-satellite point of a satellite with orbital inclination i_s.

3 Determination of antenna gain

The relationship $\varphi(\alpha)$ is used to derive a function for the horizon antenna gain (dBi), $G(\varphi)$ as a function of the azimuth α, by using the actual earth station antenna pattern, or a formula giving a good approximation. For example, in cases where the ratio between the antenna diameter and the wavelength is equal to or greater than 35, the following equation is used:

$$G(\varphi) = \begin{cases} G_{amax} - 2.5 \times 10^{-3} \left(\dfrac{D}{\lambda} \varphi \right)^2 & \text{for } 0 < \varphi < \varphi_m \\ G_1 & \text{for } \varphi_m \le \varphi < \varphi_r \\ 29 - 25 \log \varphi & \text{for } \varphi_r \le \varphi < 36° \\ -10 & \text{for } 36° \le \varphi \le 180° \end{cases} \tag{97}$$

$$G_1 = \begin{cases} -1 + 15 \log \left(D/\lambda \right) \text{ dBi} & \text{for } D/\lambda \ge 100 \\ -21 + 25 \log \left(D/\lambda \right) \text{ dBi} & \text{for } 35 \le D/\lambda < 100 \end{cases}$$

$$\varphi_m = \frac{20\,\lambda}{D}\sqrt{G_{amax} - G_1} \quad \text{degrees}$$

$$\varphi_r = \begin{cases} 15.85\left(D/\lambda\right)^{-0.6} & \text{degrees} \quad \text{for} \quad D/\lambda \geq 100 \\ 100\left(\lambda/D\right) & \text{degrees} \quad \text{for} \quad 35 \leq D/\lambda < 100 \end{cases}$$

Where a better representation of the actual antenna pattern is available, it may be used.

In cases where D/λ is not given, it may be estimated from the expression:

$$20\log\frac{D}{\lambda} \approx G_{amax} - 7.7$$

where:

G_{amax}: main beam axis antenna gain (dBi)

D: antenna diameter (m)

λ: wavelength (m)

G_1: gain of the first side lobe (dBi).

ANNEX 4

Antenna gain toward the horizon for an earth station operating with non-geostationary space stations

This Annex presents methods which may be used to determine the antenna gain towards the horizon for an earth station operating to non-geostationary satellites using the TIG method described in § 2.2 of the main body of this Appendix.

1 Determination of the horizon antenna gain

In its simplest implementation, the TIG method depends on the minimum elevation angle of the beam axis of the earth station antenna (ε_{sys}), which is a system parameter that has the same value on all azimuths from the earth station. If the horizon elevation angle at an azimuth under consideration is ε_h (degrees), the minimum separation angle from the horizon at this azimuth to any possible pointing angle for the main beam axis of the antenna (φ_{min}) is equal to the difference between these two angles ($\varepsilon_{sys} - \varepsilon_h$), but it is not less than zero degrees. The maximum separation angle from the horizon at this azimuth to any possible pointing angle for the main beam axis of the antenna (φ_{max}) is equal to the difference between the sum of these two angles and 180° ($180 - \varepsilon_{sys} - \varepsilon_h$). The maximum and minimum values of horizon gain for the azimuth under consideration are obtained from the gain pattern of the earth station antenna at these off-axis angles. Where no pattern is available the pattern of § 3 of Annex 3 may be used.